Introduction Project



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Master Smart Systems Engineering
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Introduction Project

- **Goal**: bring all students to a common level on a variety of topics
  - Improve your skills (mainly Python and sensors)
  - Share your knowledge with the others
  - Create personal learning goals
- You will work in groups of 3 students (already defined)
- **Main task**: design and implement Python code to make a robot navigate a maze using data from its IR and LiDAR sensors.



Assessment

- Individual (uploaded to Brightspace, 1 page, *not* graded):
  - set **personal learning goals (20-09)** and a plan on how to achieve them
  - **reflect (02-11)** on the previously set personal learning goals
- In groups:
  - **progress presentation (29-09)**: show your colleagues and teachers your progress (*not* graded).
  - **Demonstration (20-10)**: live demonstration of your robot navigating the maze (graded)
  - **Report/paper (25-10)**; IEEE format, max. 6 pages (graded)

Teachers

- Project tutors:
  - Ewout Bergsma (e.bergsma@pl.hanze.nl - BEEO) – groups 5-7
  - Felipe N. Martins (fe.nascimento.martins@pl.hanze.nl - NASF) – groups 1-4

- Personal learning goals:
  - Francisco (Paco) Guzman (f.j.guzman.munoz@pl.hanze.nl - GUFO)

Groups

Group	Student number	Family name	Name	Email Hanze
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Details

- Maze will be unknown to the robots
- There will be no bridges (ground is flat)
- The maze can have loops
- Walls will be at an angle of 90 degrees from each other (approximately)
- Completing the maze means the robot goes from start to finish (pre-defined positions).
- Each group gets 8 + 8 minutes (disjointed) to try completing the task.

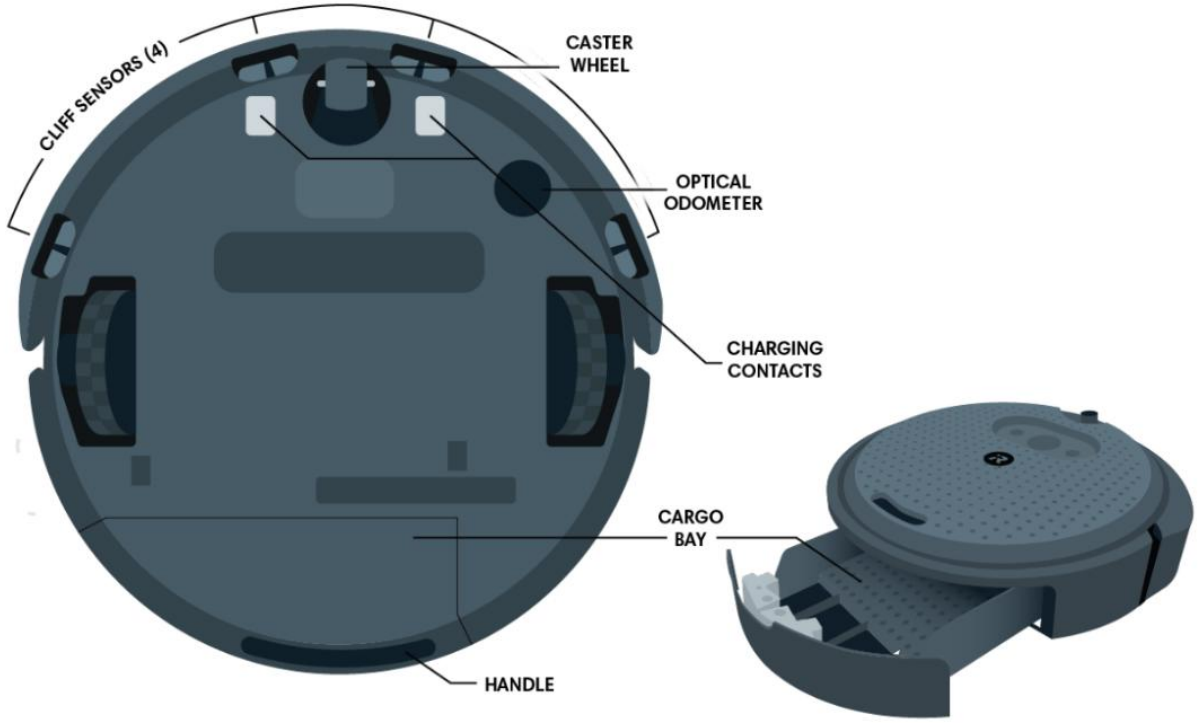


iRobot Create 3

https://iroboteducation.github.io/create3_docs/hw/overview/

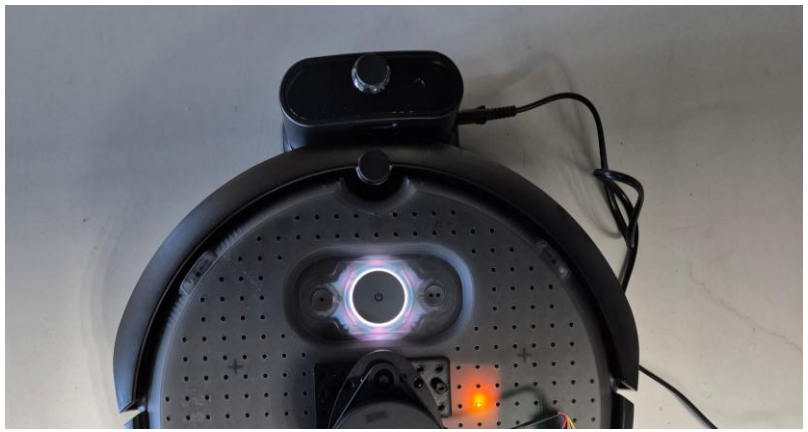
https://python.irobot.com/





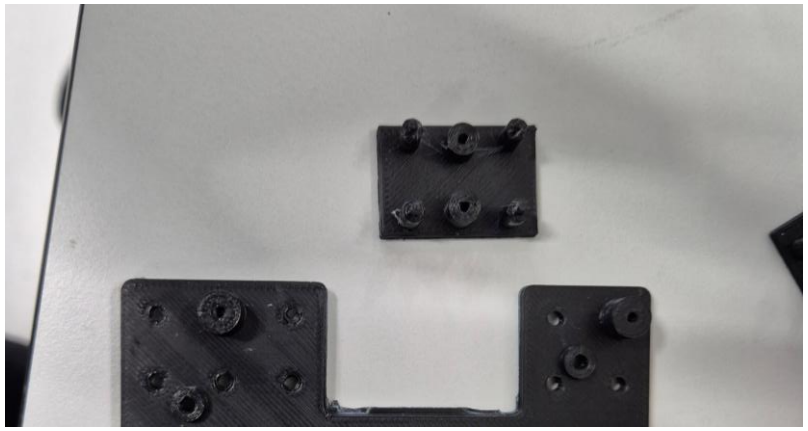












You will install the LiDAR!

Questions?

